Multiuser Capacity in Block Fading with no Channel State Information

SHLOMO SHAMAI (SHITZ)*

Department of Electrical Engineering
Technion - Israel Institute of Technology
Haifa 32000
Israel
sshlomo@ee.technion.ac.il

THOMAS L. MARZETTA

Mathematical Sciences Research Center
Bell Laboratories, Lucent Technologies
600 Mountain Avenue
Murray Hill, NJ 07974
USA
tlm@research.bell-labs.com

Abstract

Consider M independent users, each user having his own transmit antenna, that transmit simultaneously to a receiver equipped with N antennas through a Rayleigh block-fading channel having a coherence interval of T symbols, with no channel state information available to either the transmitters or to the receiver. The total transmitted power is independent of the number of users. For a given coherence time T, we wish to identify the best multi-access strategy that maximizes the total throughput.

If perfect channel state information were available to the receiver, it is known that the total capacity increases monotonically with the number of users. If the channel state information is available to both the receiver and to all transmitters, the throughput maximizing strategy implies for $N = 1$ that only the single user who enjoys the best channel condition transmits. In the absence of channel state information one is forced to a radically different conclusion. In particular we show that if the propagation coefficients take on new independent values for every symbol (e.g., $T = 1$) then the total capacity for any $M > 1$ users is equal to the capacity for $M = 1$ user, in which case TDMA is an optimal scheme for handling multiple users. This result follows directly from a recent treatment of the single-user multiple antenna block-fading channel.

Again, motivated by the single-user results, one is lead to the following conjecture for the multiple user case: for any $T > 1$ the maximum total capacity can be achieved by no more than $M = T$ users. The conjecture is supported by establishing the asymptotic result that, for a fixed N and a constant M/T for large T, the total capacity is maximized when $M/T \to 0$, which yields a total capacity per symbol of $N \log(1 + \rho)$, where ρ is the expected SNR at the receiver. We further support the conjecture by examining the asymptotic behavior with large ρ for fixed M, T and $N \leq T$.

Index Terms: wireless communication, multiple users, Rayleigh block-fading, multi-element antenna arrays.

*This research was performed, in part, while the author was visiting the Mathematical Sciences Research Center, Bell Laboratories, Lucent Technologies

1 Introduction

Multiple antenna wireless links are of ever-increasing interest to the research community, undoubtedly because of the spectacular throughputs that are obtainable under certain conditions [3, 11]. For a single-antenna link with no channel state information available at the transmitter, fading is never beneficial. On the contrary, the potential benefits of multiple antennas can only be realized in a fading environment. Our interest is the flat-fading regime where, over the frequency support of the transmitted signals, the channel response is approximately constant.

Throughout we shall adopt a block-fading model [10], where the channel propagation matrix is constant for T symbol periods, after which it jumps to a new independent value for another T symbols, and so on. In general our goal is to maximize throughput, in the Shannon sense, implying that coding is performed over many independent T-symbol coherence intervals. We will not consider performance within a single coherence interval, where Shannon coding cannot be performed, and which requires notions such as outage capacity [10], delay-limited capacity [4], or space-time autocoding [6].

A detailed survey of information theory for fading channels is given in [2]. For our purposes it is convenient to subdivide the multiple-antenna problem area as shown in Table 1, with a limited number of references provided for the sub-areas.

	CSI at transmitter & receiver	CSI at receiver	no CSI
single user	[11]	[3, 11]	[1, 9, 14]
multi-user	[7]	[11]	

Table 1: Wireless links require special treatment, depending on the number of users, and the amount of channel state information (CSI) that is available.

Usually channel state information is acquired by the transmission of a set of known training signals, with the required symbol duration of the training signals proportional to the number of transmit antennas. The overhead that is represented by training can be considerable. In fact, where one is free to choose the number of antennas to maximize the net throughput for a fixed coherence interval, half of the coherence interval is used for training [8]. Training, however, is not always inefficient. See [14, 5], which demonstrate that training may preserve basic "degrees of freedom" in the system. There certainly is no fundamental mandate to perform

training, so the assumption of no CSI is the most natural one to make.

For a single user, the assumption of no CSI at the receiver versus the assumption that

the receiver has CSI changes the problem drastically. With CSI at the receiver only, the capacity attaining signals are independent among the transmit antennas, and capacity increases linearly with the smaller of the number of transmit and receive antennas. Without CSI, the capacity attaining signals are temporally orthogonal among the transmit antennas within the coherence interval, and there is no point in making the number of transmit

antennas M greater than the symbol duration of the coherence interval T. When CSI is available to both the transmitter and the receiver, the single user can further improve performance by water-pouring optimization, which also introduces signal correlation among the antennas.

In contrast to the single user, multiple users must transmit independent signals. When CSI is available to the receiver alone, the multiple users can collectively duplicate the throughput of a single user, whose optimal signals are independent among the transmit antennas. With CSI available to both transmitter and receiver (and for the case of a single receive antenna, $N = 1$), maximum throughput is achieved by a strategy called "controlled TDMA" [2], where only the user who possesses the strongest propagation coefficient transmits during each coherence interval. This specific result does not necessarily hold for $N > 1$. The case of CSI available either to the receiver, or to the transmitters and receiver, is equivalent to a coded DS-CDMA system where the channel coefficients play the role of signature sequences of length N (processing gain), where M stands for the number of users [12].

The focus of this note is the case where no CSI is available to either the transmitters or to the receiver. Here we conjecture that maximum throughput can be achieved when the number of users is less than or equal to the symbol duration of the coherence interval, and we establish the validity of this conjecture for three special cases.

2 Signal Model

We use a block-fading model, with coherence interval T [10, 2, 9] where M independent users simultaneously transmit to a single receiver equipped with N antennas in a flat-fading environment, where each user has sole access to one of M transmit antennas, and where nobody has any channel

state information. The fading is described by a $M \times N$ complex-valued propagation vector H, which remains constant for T symbol periods, after which it jumps to a new independent value for another T symbol, and so on. During a coherence interval the M users collectively transmit a $T \times M$ complex matrix S, whose columns are statistically independent, and the receiver records a $T \times N$ complex matrix X,

$$X = \sqrt{\frac{\rho}{M}} SH + W \,, \tag{1.1}$$

where W is a $T \times N$ vector of additive receiver noise, whose components are independent, zero-mean complex Gaussian with unit variance (CN(0, 1)). The M components of H are assumed independent and zero-mean, with unit-variance. The independence of the columns of S implies that the users are acting with no cooperation among themselves. We enforce an expected power constraint

$$\mathrm{tr}\,\mathrm{E}\left\{SS^{\dagger}\right\} = T \cdot M \,. \tag{1.2}$$

This constraint, when combined with the normalization $1/\sqrt{M}$ in (1.1), implies that the total transmitted power remains constant as the number of users changes, and that ρ represents the expected SNR at each receiver antenna.

We wish to choose the joint probability density of the components of S, subject to the independence of its M columns, and subject to the power constraint (1.2), to maximize the mutual information with no CSI available to anyone (receiver and transmitters),

$$I(X;S) \triangleq \mathrm{E}\left\{\log\left(\frac{p(X|S)}{p(X)}\right)\right\} \tag{1.3}$$

This maximization yields the total throughput or capacity.

It is convenient to assume Rayleigh fading, where the independent components of H are distributed as CN(0, 1), however this assumption can be relaxed for some of our asymptotic results. For Rayleigh fading the conditional density takes the form

$$p(X \mid S) = \frac{\exp\left(-X^{\dagger}\Lambda^{-1}X\right)}{\pi^{TN}\det^{N}\Lambda} \,, \tag{1.4}$$

where $\Lambda = I_T + (\rho/M)\,SS^{\dagger}$ and I_T denotes the $T \times T$ identity matrix.

3 Upper Bound on Capacity; Capacity for T = 1

The constraint that the columns of S be statistically independent makes the maximization of the mutual information (1.3) difficult. If we relaxed this constraint and permitted dependent columns we would obtain an upper bound on the multiuser capacity, which is the cooperative multiple-user capacity equaling the capacity for a single user who has access to M transmit antennas. This problem was studied for the case $T = M = 1$ by [1] and for general T, N and M by [9], with the following results:

1. The capacity for $M > T$ is equal to the capacity for $M = T$;

2. In choosing $p(S)$ to maximize mutual information, the transmitted matrix can, without penalty, be constrained to have the factorization

$$S = \Phi V \,, \tag{2.1}$$

   where Φ and V are statistically independent, Φ is an isotropically random $T \times T$ unitary matrix (i.e., the columns are orthonormal, and the joint probability density of the elements of Φ is unchanged if Φ is premultiplied by any deterministic unitary matrix), and V is real, nonnegative, $T \times M$ diagonal.

Note that for the capacity-attaining signal, the columns of S are *not* independent. In contrast, if perfect CSI were available to the receiver [11] the components of S would be independent CN$(0, 1)$, and the capacity would increase monotonically with M.

If perfect CSI were available to the receiver, the total M-user capacity would be equal to the single-user/M-antenna capacity [11]. If CSI were available to both receiver and transmitter, and if $N = 1$, only one user should transmit in each block: that who enjoys the best channel condition [7]. This strategy was referred to as controlled random TDMA in [2]. This strategy does not necessarily scale with N for an arbitrary propagation matrix H. This can be deduced by examining H matrices that satisfy the Welch bound, that is $\frac{1}{M} H^\dagger H = I_N$, and $\frac{1}{N} H H^\dagger$ has diagonals equal to one. In this case it can be shown [13], via the analogy to coded DS-CDMA previously mentioned, that capacity equal to $N \log(1 + \rho)$ is achieved only if all $M \geq N$ users transmit i.i.d. equal-power signals.

We conclude that the absence of CSI drastically changes the problem, and that the total M-user capacity is generally less than the single-user/M-antenna capacity.

For the special case where $T = 1$ (arbitrarily fast fading) a single user having M antennas can attain the same performance with a single antenna, so the total capacity for M users is equal to the capacity for one user. This establishes the optimality of TDMA for $T = 1$ and unavailable CSI. This conclusion extends also to all fading matrices H which are statistically invariant under a unitary transformation.

For the general case $T > 1$ an obvious conjecture [2], in light of the results cited from [9], is that the total capacity for any $M > T$ is equal to the total capacity for $M \leq T$. At present we are unable to prove this conjecture, but in the following sections we make some headway by studying the case where T and M grow big simultaneously, or ρ grows asymptotically large.

4 Asymptotic Results for Large T, M

We begin by deriving some upper and lower bounds on the mutual information (1.3). Throughout, we only assume that the components of H are independent, zero-mean, and unit variance. The central relation is

$$\begin{aligned} I(X;S) &= \mathrm{E}\left\{\log\left[\frac{p(X|S)\cdot p(X|S,H)}{p(X)\cdot p(X|S,H)}\right]\right\} \\ &= \mathrm{E}\left\{\log\left[\frac{p(X|S,H)}{p(X)}\right]\right\} - \mathrm{E}\left\{\log\left[\frac{p(X|S,H)}{p(X|S)}\right]\right\} \\ &= I(X;S,H) - I(X;H|S)\,. \end{aligned} \tag{3.1}$$

The mutual information $I(X;S,H)$ corresponds to an additive Gaussian noise channel, $X = (\rho/M)^{1/2}Z + W$, where $Z = SH$. This mutual information is maximized by making Z zero-mean complex Gaussian with independent columns, with covariance of each column equal to that of each column of SH, or $\mathrm{E}\left\{SS^{\dagger}\right\}$. This bound, combined with (3.1), gives an upper bound on $I(X;S)$,

$$\begin{aligned} I(X;S) &< I(X;S,H) \\ &< N\,\log\det\left(I_T + \frac{\rho}{M}\mathrm{E}\left\{SS^{\dagger}\right\}\right) \end{aligned} \tag{3.2}$$

Subject to the power constraint (1.2), this is maximized for $\mathrm{E}\left\{SS^{\dagger}\right\} = M \cdot I_T$, which gives an upper bound,

$$I(X;S) \leq TN \log(1+\rho) \,. \tag{3.3}$$

Now we obtain a lower bound on the first term in (3.1),

$$\begin{aligned} I(X;S,H) &= \mathrm{E}\left\{\log\left[\frac{p(X|S,H)\cdot p(X|H)}{p(X)\cdot p(X|H)}\right]\right\} \\ &= \mathrm{E}\left\{\log\left[\frac{p(X|S,H)}{p(X|H)}\right]\right\} + \mathrm{E}\left\{\log\left[\frac{p(X|H)}{p(X)}\right]\right\} \\ &= I(X;S|H) + I(X;H) \\ &\geq I(X;S|H) \,. \end{aligned} \tag{3.4}$$

We attempt to maximize $I(X;S|H)$,

$$I(X;S|H) = \mathrm{E}_H\left\{\mathrm{E}\left\{\log\left[\frac{p(X|S,H)}{p(X|H)}\right]|H\right\}\right\} \,. \tag{3.5}$$

This is done by choosing S to be zero-mean complex Gaussian, with independent components having equal variance [11], which yields

$$I(X;S|H) = T \cdot \mathrm{E}\left\{\log\det\left(I_N + \frac{\rho}{M}H^{\dagger}H\right)\right\} \,. \tag{3.6}$$

Next we obtain an *upper* bound on the second term in (3.1),

$$I(X;H|S) = \mathrm{E}_S\left\{\mathrm{E}\left\{\log\left[\frac{p(X|S,H)}{p(X|S)}\right]|S\right\}\right\} \,. \tag{3.7}$$

This mutual information corresponds to the fictitious case of a user sending the signal H through a random propagation matrix S, where S is known at the receiver. The inner conditional mutual information is upper bounded when X is zero-mean complex Gaussian; this is achieved by making the elements of H independent $\mathrm{CN}(0,1)$, which gives

$$I(X;H|S) \leq N \cdot \mathrm{E}\left\{\log\det\left(I_T + \frac{\rho}{M}SS^{\dagger}\right)\right\} \,. \tag{3.8}$$

Note that in the process of maximizing (3.5), S has been made white Gaussian. The combination of (3.1), (3.4), (3.6), and (3.8) gives a lower bound on mutual information,

$$I(X;S) \geq T \cdot \mathrm{E}\left\{\log\det\left(I_N + \frac{\rho}{M}H^\dagger H\right)\right\} - N \cdot \mathrm{E}\left\{\log\det\left(I_T + \frac{\rho}{M}SS^\dagger\right)\right\} . \tag{3.9}$$

We use a common matrix identity and Jensen's inequality as follows, to bound the second term, with the components of S independent, $\mathrm{CN}(0,1)$,

$$\begin{aligned}
\mathrm{E}\left\{\log\det\left(I_T + \frac{\rho}{M}SS^\dagger\right)\right\} &= \mathrm{E}\left\{\log\det\left(I_M + \frac{\rho}{M}S^\dagger S\right)\right\} \\
&< \log\det\left(I_M + \frac{\rho}{M}\mathrm{E}\left\{S^\dagger S\right\}\right) \\
&= \log\det\left(I_M + \frac{\rho T}{M}I_M\right) \\
&= M\log\left(1 + \frac{\rho T}{M}\right) .
\end{aligned} \tag{3.10}$$

See also the symmetry properties in [11, Remark 3]. The substitution of (3.10) into (3.9) gives

$$I(X;S) \geq T \cdot \left[\mathrm{E}\left\{\log\det\left(I_N + \frac{\rho}{M}H^\dagger H\right)\right\} - \frac{MN}{T}\log\left(1 + \frac{\rho T}{M}\right)\right] \tag{3.11}$$

We simultaneously let T and M become big while maintaining a constant ratio M/T. This gives [11]

$$I(X;S) \geq T \cdot N\left[\log(1+\rho) - \frac{M}{T}\log\left(1 + \frac{\rho T}{M}\right)\right] \tag{3.12}$$

Finally we let the ratio grow small, $M/T \to 0$, which yields the asymptotic result

$$I(X;S) \geq TN\log(1+\rho) . \tag{3.13}$$

Together, (3.3) and (3.13) imply that for fixed N and asymptotically, for large T and M, and with $M/T \to 0$, the mutual information is $I(X;S) = TN\log(1+\rho)$, which is equal to the capacity for a single user, having an unlimited number of transmit antennas with perfect CSI at the receiver. This supports our original conjecture (e.g., mutual information maximized for $M/T \leq 1$).

5 Asymptotics with Large ρ

To support further our conjecture we now focus on the case where T, M and N are fixed, while the signal-to-noise ratio $\rho \to \infty$. This asymptotically high SNR region is appropriate for multiple users operating

in an extremely high data rate regime. Again, resorting to an upper bound by allowing the full cooperation at the transmitter sites we invoke recent results by Zheng & Tse [14] characterizing the asymptotic capacity of the single user M-transmit, N-receiver systems with coherence time T.

The capacity behaviour [14] is

$$\frac{C(M,N,T;\rho)}{\log \rho} \underset{\rho\to\infty}{\longrightarrow} TM^* \left(1 - \frac{M^*}{T}\right) , \tag{4.1}$$

where

$$M^* = \min \left\{ M, N, \left\lfloor \frac{T}{2} \right\rfloor \right\} \tag{4.2}$$

Here $C(M,N,T;\rho)$ stands for the capacity over the coherence interval T (not normalized) with M transmit and N receive antennas, and SNR $= \rho$. With $N = 1$, this result supports our conjecture, as immaterial of what the coherence time T is (fixed though), having a single operating transmitter yields the optimal possible asymptotics with respect to ρ.

Clearly, we cannot infer from [14] the optimal strategy for $N > 1$. For $N < \left\lfloor \frac{T}{2} \right\rfloor$, the fact that the single-user capacity does not require more than N transmitters doesn't imply necessarily the same conclusion in the multiple user case. This is because, in the single user case, capacity achieving signals are in general correlated for different transmitting antennas, and this cannot be the case for multiple users.

To investigate the behavior for $N > 1$, though $N < T$, we resort again to independent Gaussian signals, and we explicitly assume that the components of H are independent Gaussian. We denote this by a subscript G. We find

$$I_G(X;S) \geq I_G(X\,;\,S|H) - I_G(X\,;\,H|S) \tag{4.3}$$

Notice that

$$I_G(X\,;\,S|H) = T \cdot \mathcal{C}_{\mathrm{SI}}(M,N;\rho) , \tag{4.4}$$

where $\mathcal{C}_{\mathrm{SI}}(M, N; \rho)$ stands for the single user normalized capacity per unit time with M transmitters and N receiving antennas and signal-to-noise ratio ρ [11]. The symmetry of the signal model (1.1) yields,

$$I_G(X\,;\,H|S) = N \cdot \mathcal{C}_{\mathrm{SI}}(M, T\,;\,\rho)\,. \tag{4.5}$$

Symmetry considerations [11] imply that

$$\mathcal{C}_{\mathrm{SI}}(a, b; b\rho) = \mathcal{C}_{\mathrm{SI}}(b, a; a\rho) \tag{4.6}$$

Now, it is known [11] that

$$\frac{\mathcal{C}_{\mathrm{SI}}(M, N\,;\,\rho)}{\log \rho} \underset{\rho\to\infty}{\longrightarrow} \min(M, N)\,, \tag{4.7}$$

which when combined with (4.3) yields

$$\frac{I_G(X\,;\,S)}{\log \rho} \underset{\rho\to\infty}{\longrightarrow} \geq T \min(M, N) - N \min(M, T)\,. \tag{4.8}$$

Consider now the case of fixed N and T and $N \leq T$. Clearly, the asymptotic expression is optimized by letting $M = N$, which yields,

$$\frac{I_G(X\,;\,S)}{\log \rho} \underset{\rho\to\infty}{\longrightarrow} \geq TN\left(1 - \frac{N}{T}\right)\,. \tag{4.9}$$

Note that for $2N \leq T$, this yields the optimal performance as deduced by (4.1), (4.2). For $\left\lfloor \frac{T}{2} \right\rfloor < N \leq T$, there is a gap between the behavior of (4.9) (where $M = N$) and (4.1) which is optimized for $M = \left\lfloor \frac{T}{2} \right\rfloor < N$. We conjecture that (4.9) reflects the actual asymptotic behavior in this case and the penalty with respect to (4.1) is attributed to the independent signaling at each transmitting antenna.

6 Conclusions

The absence of CSI drastically alters the behavior of a wireless link for independent multiusers.

We conjecture that, from the standpoint of maximizing throughput, the number of users should not exceed the symbol duration of the coherence interval. We have established the correctness of the conjecture for three special cases: 1) a coherence interval comprising one symbol, 2) a coherence interval of unlimited duration, and 3) an unlimited SNR.

As mentioned in the introduction, there is an interesting analogy between multiple user, multiple antenna links and coded DS-CDMA with random signatures. In the regime considered here, the signatures are chosen in an absolutely random fashion for every T-symbol coherence interval, without their values being known by any of the receivers, whether legitimate or not. It is not surprising that our results also conform with the autocoding capacity [6], where, for the case of unknown CSI the number of actively operating users at each time instant is small compared with T.

References

[1] I. C. Abou Faycal, M. D. Trott and S. Shamai (Shitz), "The Capacity of Discrete-Time Rayleigh Fading Channels," in *Proc. Int. Symp. Info. Th.*, Ulm, Germany, 1997, p. 473; also, to appear in *IEEE Trans. on Information Th.*.

[2] E. Biglieri, J. Proakis, and S. Shamai (Shitz), "Fading Channels: Information-Theoretic and Communications Aspects", *IEEE Trans. Information Th.*, vol 44, no. 6, pp. 2619–2992, October 1998.

[3] G. J. Foschini, "Layered Space-Time Architecture for Wireless Communication in a Fading Environment when Using Multi-Element Antennas", *Bell Labs. Tech. J.*, vol. 1, no. 2, pp. 41-59, 1996.

[4] S. V. Hanly and D. N. Tse, "The Multi-access Fading Channel: Shannon and Delay Limited Capacities", *Proc. 33d Allerton Conf.*, Monticello, IL, p. 786-795, 1995.

[5] B. Hassibi and B. M. Hochwald, "How Much Training is Needed in Multiple-Antenna Wireless Links", *submitted to IEEE Trans. Information Th.*.

[6] B. M. Hochwald, T. L. Marzetta, and B. Hassibi, "Space-Time Autocoding", *submitted to IEEE Trans. Information Th.*; also, "Space-Time Autocoding: Arbitrarily Reliable Communication in a Single Fading Interval", *Proc. 2000 IEEE Int. Symp. on Information Th.*, p. 313, Sorrento, Italy, 25-30 June, 2000..

[7] R. Knopp and P. A. Humblet, "Information Capacity and Power Control in Single-Cell Multiuser Communications", in *Proc. Int. Conf. Communications, ICC'95*, pp. 331–335, Seattle, WA, USA, June 18–22, 1995.

[8] T. L. Marzetta, "BLAST Training: Estimating Channel Characteristics for High-Capacity Space-Time Wireless", *Proc. 37th Annual Allerton Conference on Communications, Control, and Computing*, pp. 958-966, Monticello, IL, Sept. 22-24, 1999.

[9] T. L. Marzetta and B. M. Hochwald, "Capacity of a Mobile Multiple-Antenna Communication Link in Rayleigh Flat Fading", *IEEE Trans. on Information Th.*, vol. 45, no. 1, pp. 139–157, January 1999.

[10] L. H. Ozarow, S. Shamai (Shitz) and A. D. Wyner, "Information Theoretic Considerations for Cellular Mobile Radio", *IEEE Trans. Veh. Tech.*, vol. 43, pp. 359–378, May 1994.

[11] I. E. Telatar, "Capacity of Multi-Antenna Gaussian Channels, *European Transactions of Telecommunications (ETT)*, Vol. 10, No. 6, pp. 585–596, November/December 1999.

[12] S. Verdu and S. Shamai (Shitz), "Spectral Efficiency of CDMA with Random Spreading", *IEEE Trans. on Information Th.*, vol. 45, no. 2, pp. 622-640, March 1999.

[13] P. Viswanath and V. Anantharam, "Optimal Sequences and Sum Capacity of Synchronous CDMA Systems", *IEEE Trans. Information Th.*, vol. 45, no. 6, pp. 1984-1991, Sept. 1999.

[14] L. Zheng and D.N.C. Tse, "Packing Spheres in the Grassmann Manifold: A Geometric Approach to the Non-Coherent Multi-Antenna Channel," preprint, April 30, 2000. See also: ISIT'2000, p. 364, Sorrento, Italy, 25–30 June, 2000.

The *Center for Communication and Information Technologies* (CCIT)

is managed by the Department of Electrical Engineering.

This Technical Report is listed also as

EE PUB #1258, August 2000.